



Robust adaptive attenuation of unknown periodic disturbances in uncertain multi-input multi-output systems[☆]



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ABSTRACT

In high-performance high-accuracy systems, the attenuation of vibrational disturbances is essential. In this paper, we design and analyze a robust output-feedback adaptive control scheme to attenuate noise-corrupted vibrational disturbances with unknown characteristics for multi-input multi-output linear time-invariant systems in the presence of plant unstructured modeling uncertainties. The conditions and the design parameters that contribute to the performance of the closed-loop system are investigated. The main ingredients of the proposed scheme include the use of a compensator to properly shape the singular values of the plant model, a robust adaptive law to handle unmodeled dynamics and output random noise, as well as an over-parametrization in the estimated parameters in order to provide flexibility for performance improvement. Analysis and simulations together with practical design considerations are presented to demonstrate the efficacy of the proposed scheme.

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1. Introduction

Vibrations and unwanted sound suppression is a key enabling technology in a vast array of aerospace and industrial control applications. In many defense, industrial, and medical applications, unwanted sound and vibrations primarily caused by rotating components are dominantly periodic disturbances (Benesty, Sondhi, & Huang, 2008; Elliott & Nelson, 1993; Nelson & Elliott, 1992; Rudd, Lim, Li, & Lee, 2012). For example, high energy laser and optical communication systems can be very sensitive to platform vibrations and other jitter sources. Feedback adaptive control techniques have been proven to be quite effective in attenuating unknown periodic disturbances and enhancing performance (Perez-Arancibia, Gibson, & Tsao, 2009). Their application to advanced flight systems such as spacecraft (Lau, Joshi, Agrawal, & Kim, 2006), turboprop aircraft, and helicopters (Patt, Liu, Chandrasekar, Bernstein, & Friedmann, 2005) has also motivated a great deal of research in recent years (Aranovskiy & Freidovich, 2013; Basturk & Krstic, 2014; Bodson, 2005; Chen & Tomizuka, 2012; Isidori, Marconi, & Praly, 2012; Jafari, Ioannou, Fitzpatrick, & Wang,

2014; Kim, Kim, Chung, & Tomizuka, 2011; Landau, Alam, Martinez, & Buche, 2011; Landau, Alma, Constantinescu, Martinez, & Noe, 2011; Marino & Tomei, 2011, 2013; Nikiforov, 2004; Pigg & Bodson, 2010). The aforementioned papers are mainly for rejection of unknown periodic disturbances for single-input single-output (SISO) systems. For multi-input multi-output (MIMO) systems with strong coupling between the channels, however, the results for SISO systems are not directly applicable. In Pulido, Toledo, and Loukianov (2011), an estimator is designed to estimate the parameters of an unknown disturbance with a known number of distinct frequencies, and an adaptive internal-model based controller is then designed to suppress the effect of disturbances on the output of a MIMO plant. The limitations of internal-model based rejection of periodic disturbances with known time-varying frequencies have been studied in Kinney and de Callafon (2011). Recently, in Houtzager, van Wingerden, and Verhaegen (2013), a control method for MIMO systems has been proposed for rejection of periodic wind disturbances in wind turbines. These results are mostly for pure sinusoidal disturbances without considering the effect of low-amplitude broadband noise on the output of the plant. Moreover, it is mainly assumed that the plant model is perfectly known, without considering the effect of inevitable modeling uncertainties. The authors in Jafari and Ioannou (2013) proposed an adaptive scheme for rejection of unknown periodic disturbances with constant parametric characteristics for MIMO systems in the absence of plant modeling error. It is well known in robust adaptive control (Anderson, 2005; Ioannou & Sun, 1996) that adaptive

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feedback systems designed for perfectly modeled plants could easily go unstable in the presence of small-size plant modeling uncertainties.

This paper presents the design and analysis of a robust adaptive controller for the attenuation of noise-corrupted vibrational disturbances with unknown characteristics in the presence of unstructured modeling uncertainties. In addition, the paper analyzes the trade-offs between performance and robustness which are used to choose certain design parameters. The main contributions of this paper are: (i) design and analysis of an output feedback adaptive controller for linear MIMO plants which is robust with respect to unmodeled dynamics, and additive output noise, (ii) design a pre-compensator to shape the singular values of the plant to have a high enough gain over the expected frequency range of the disturbance and low enough gain over the frequencies where the modeling error may be dominant. These modifications are beneficial to both performance and robustness with respect to noise and high-frequency unmodeled dynamics.

This paper is organized as follows: Section 2 presents some preliminaries and the notation used throughout the paper. The problem is defined and formulated in Section 3. Sections 4 and 5 contain the main results of the paper. The effectiveness of the proposed adaptive scheme is demonstrated in Section 6 using a simulation example. The concluding remarks are summarized in Section 7.

2. Preliminaries

An n_u -input, n_y -output finite-dimensional linear time-invariant system with real-rational transfer function matrix $H(z)$, input vector $u(k)$, and output vector $y(k)$ may be expressed as $y(k) = H(z)[u(k)]$, where $z = e^{j\omega T_s}$ is the operator of the z -transformation and T_s is the sampling period. For a signal x , the truncated signal x_k is defined as $x_k(i) = x(i)$, for $0 \leq i \leq k$ and $x_k(i) = 0$, for $i > k$. We use the $\mathcal{H}_\infty$ -norm $\|H(z)\|_\infty = \max_{\theta \in [0, \pi]} \sigma_{\max}(H(e^{j\theta}))$, where $\sigma_{\max}$ denotes the maximum singular value, and the $\mathcal{L}_1$ -norm $\|H(z)\|_1 = \max_{i=1}^{n_y} \sum_{j=1}^{n_u} \|h_{ij}\|_1$, where h_{ij} is the impulse response of the ij -element of $H(z)$, where $\|h_{ij}\|_1 = \sum_{k=0}^{\infty} |h_{ij}(k)|$ (Ioannou & Fidan, 2006; Ioannou & Sun, 1996; Skogestad & Postlethwaite, 2005). The $\mathcal{H}_\infty$ norm and the $\mathcal{L}_1$ norm are induced norms and satisfy the multiplicative property. The two norms are related as $\|H\|_\infty \leq \sqrt{n_y} \|H\|_1 \leq \sqrt{n_y n_u} (2n + 1) \|H\|_\infty$, where n is the degree of a minimal realization of the corresponding system (Dahleh & Diaz-Bobillo, 1995).

The Swapping Lemma (Ioannou & Fidan, 2006) states that for signal vectors $x(k)$, $y(k)$ and a proper stable rational scalar transfer function $H(z)$ with a minimal realization (A, B, C, D) , i.e., $H(z) = C^T(zI - A)^{-1}B + D$, we have

$$H(z)[x^T(k)y(k)] = x^T(k)H(z)[y(k)] + H^c(z)[(H^b(z)[y^T(k)])\Delta x(k)],$$

where $\Delta x(k) = x(k+1) - x(k)$, $H^c(z) = -C^T(zI - A)^{-1}$, and $H^b(z) = z(zI - A)^{-1}B$.

We say that the signal x is μ -small in the mean square sense and write $x \in \mathcal{S}(\mu)$, when $\sum_{i=k}^{k+N-1} x(i)^T x(i) \leq c_1 N \mu + c_2$, $\forall k, N$, and some constant $c_1, c_2 \geq 0$ (Ioannou & Fidan, 2006; Ioannou & Sun, 1996).

A proper real-rational transfer function matrix $M(z)$ analytic in $|z| \geq 1$ is called *inner* if it satisfies $M^T(z^{-1})M(z) = I$ (an inner function is square or has more rows than columns), and is called *outer* if it is right invertible with a right inverse proper and analytic in $|z| > 1$ (an outer matrix is square or has more columns than rows). Any proper real-rational transfer function matrix $H(z)$ analytic in $|z| \geq 1$ with no zeros on the unit circle has an *inner-outer factorization* $H(z) = H_{in}(z)H_{out}(z)$, where $H_{in}(z)$ is inner,

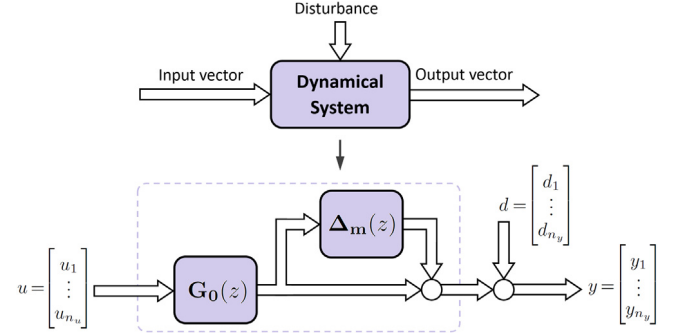


Fig. 1. The structure of the open-loop plant model with multiplicative output uncertainty.

$H_{out}(z)$ is outer, and the right inverse of $H_{out}(z)$ is proper and analytic in $|z| \geq 1$ (Ionescu & Oara, 1996; Lin, Chen, Saberi, & Shamash, 1996). A proper real-rational transfer function matrix $M(z)$ analytic in $|z| \geq 1$ is called *all-pass* if it is square and satisfies $M^T(z^{-1})M(z) = I$; clearly a square inner function is all-pass.

3. Problem formulation

Consider an uncertain multi-input multi-output plant

$$y(k) = G(z)[u(k)] + d(k) \\ = (I + \Delta_m(z))G_0(z)[u(k)] + d(k), \quad (1)$$

where $y \in \mathbb{R}^{n_y}$ is the measurable output, $u \in \mathbb{R}^{n_u}$ is the control input, $d \in \mathbb{R}^{n_y}$ is an unknown and unmeasurable bounded disturbance, $G(z) = (I + \Delta_m(z))G_0(z)$ is the $n_y \times n_u$ transfer matrix of the plant, $G_0(z) = [G_{0ij}(z)]_{n_y \times n_u}$ is the known modeled part of the plant which is bounded-input bounded-output (BIBO) stable and is possibly non-minimum phase, and $\Delta_m(z)$ is the multiplicative output uncertainty which is unknown but small in some norm sense in the low frequency range and may be relatively large at high frequencies (a qualitative assumption we later make more precise). We assume $\Delta(z) = \Delta_m(z)G_0(z)$ is proper and analytic in $|z| \geq 1$ which is a standard assumption in robust control theory for permissible perturbations. No knowledge about the structure and parameters of the unmodeled dynamics $\Delta_m(z)$ is assumed to be known. The structure of the open-loop plant model is shown in Fig. 1.

In many applications, the dominant part of the narrow-band vibrational disturbances is often modeled as a sum of sinusoidal terms with unknown frequencies, magnitude, and phases. For each output channel we assume the disturbance can be expressed as

$$d_j(k) = d_{sj}(k) + v_j(k) \\ = \sum_{i=1}^{n_{fj}} a_{ij} \sin(\omega_{ij}k + \varphi_{ij}) + v_j(k), \quad (2)$$

for $j = 1, \dots, n_y$, where the sum of the periodic terms $d_{sj}(k)$ is the dominant part of the disturbance, n_{fj} is the number of distinct frequencies, $v_j(k)$ is a zero-mean bounded noise whose amplitude is much smaller than that of $d_{sj}(k)$, and the parameters a_{ij} , ω_{ij} [rad/sample], and φ_{ij} [rad] are all unknown. The number of distinct frequencies in d_s whose corresponding amplitude is above the output noise level is bounded from above by a known bound. We express the additive output disturbance vector as

$$d(k) = d_s(k) + v(k), \quad (3)$$

where $d_s = [d_{s1}, \dots, d_{sn_y}]^T$ and $v = [v_1, \dots, v_{n_y}]^T$.

The objective is to design a control signal u to suppress the effect of d on y as much as possible. In this paper we use the controller

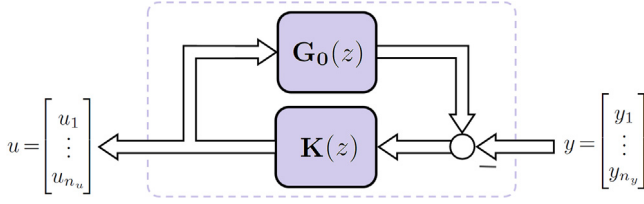


Fig. 2. The structure of the controller.

structure shown in Fig. 2 where the filter $K(z)$ is to be designed such that all signals in the closed-loop system are uniformly bounded and the norm of the output signal y is minimized.

As a first step we consider the ideal case where the frequencies of the disturbance are known exactly and no adaptation is needed. The purpose of the step is to show what can be achieved in terms of robustness and performance using the proposed controller structure when all required information about the disturbance is known. The performance and robustness margins obtained in this ideal case can be used for comparison purposes with the adaptive case where the disturbance frequencies are completely unknown. From the plant and controller structure shown in Figs. 1 and 2, the transfer matrix from d to y is

$$\begin{aligned} y(k) &= (I - G_0 K) (I + \Delta_m G_0 K)^{-1} [d(k)] \\ &= (I - G_0 K) [\bar{d}(k)] \\ &= S[\bar{d}(k)], \end{aligned} \quad (4)$$

where $S(z) \triangleq I - G_0(z)K(z)$ and

$$\bar{d}(k) = (I + \Delta_m G_0 K)^{-1} [d(k)]. \quad (5)$$

From the small gain theorem (Desoer & Vidyasagar, 1975; Ioannou & Sun, 1996) if

$$\|\Delta_m(z)G_0(z)K(z)\|_\infty < 1, \quad (6)$$

then $\bar{d}$ given in (5) is guaranteed to be a bounded disturbance which contains all frequencies of d and can be modeled as a sum of some unknown noise-corrupted sinusoidal terms as $\bar{d} = \bar{d}_s + \bar{v}$, where

$$\begin{aligned}\bar{d}_s(k) &= (I + \Delta_m G_0 K)^{-1} [d_s(k)] \\ \bar{v}(k) &= (I + \Delta_m G_0 K)^{-1} [v(k)].\end{aligned}\tag{7}$$

From (4), if $K(z)$ is such that at the frequencies of $\tilde{d}_s$, $S(z)$ has zero gain in the direction of $\tilde{d}_s$, then the controller completely rejects the periodic parts of the disturbance. However, the rejection of the periodic terms of the disturbance has to guarantee that the output broadband noise is not amplified.

We consider a transfer matrix K whose elements are Finite Impulse Response (FIR) filters of order N of the form

$$\begin{aligned} K(z, \theta) &= [K_{ij}(z, \theta_{ij})]_{n_u \times n_y}, \\ K_{ij}(z, \theta_{ij}) &= \theta_{ij}^T \alpha(z), \\ \alpha(z) &\triangleq [z^{-N}, z^{1-N}, \dots, z^{-1}]^T, \end{aligned} \quad (8)$$

where $\theta_{ij} \in \mathbb{R}^N$ is the parameter vector of the ij th element of the filter and $\theta = [\theta_{11}^T, \theta_{12}^T, \dots, \theta_{n_u n_y}^T]^T \in \mathbb{R}^{N n_u n_y}$ is the concatenation of θ_{ij} 's.

As indicated earlier, we first examine conditions under which there exists a desired parameter vector θ^* for which the periodic terms of the disturbance can be completely rejected without amplifying the output noise. In addition, we examine what additional modifications can be made to the structure of Fig. 2 in order to further improve performance. In doing so we first assume that the frequencies of the output disturbance are known leading to a non-adaptive solution and results. We then use the non-adaptive control structure in combination with a robust adaptive law to deal with the case of unknown disturbance frequencies.

4. Existence of a periodic disturbance rejection controller

In this section, we examine the existence of a filter $K(z, \theta)$ of the form (8) which rejects the periodic components of the disturbance. We show under what conditions we can find a controller parameter vector θ^* for which the periodic components of the disturbance are completely rejected, assuming the disturbance parameters are known.

Let θ^* be the desired parameter vector for which complete rejection of the periodic parts of the disturbance is possible. For a given disturbance vector $d(k)$ with frequencies ω_i 's, one may solve the system of equations $G_0(z_i)K(z_i)\vec{d}(\omega_i) = \vec{d}(\omega_i)$ for θ^* , where $z_i = e^{j\omega_i}$ and $\vec{d}(\omega_i)$ is the phasor of the sinusoidal disturbance at frequency ω_i , e.g., if $\vec{d}(k) = [a_i \sin(\omega_i k + \varphi_i), b_i \sin(\omega_i k + \psi_i)]^T$, then $\vec{d}(\omega_i) = [a_i e^{j\varphi_i}, b_i e^{j\psi_i}]^T$; however, this equation is not always solvable. The existence of a disturbance rejecting parameter vector θ^* is the subject of the following theorem. It should be noted that with such a parameter vector the stability condition (6) must be satisfied, otherwise, the closed-loop system may become unstable.

Theorem 1. Let $\omega_1, \omega_2, \dots, \omega_{n_f}$ be all distinct frequencies in the disturbance vector $\vec{d}$. There exists a θ^* such that $K(z, \theta^*)$ in Fig. 2 completely rejects periodic terms in any direction if and only if $n_y \leq n_u$, $\text{rank}(G_0(e^{j\omega_i})) = n_y$, for $i = 1, 2, \dots, n_f$, and $N \geq 2n_f$. If $N = 2n_f$, the value of θ^* for complete rejection of the periodic disturbance terms in every direction is unique.

Proof. Rejection of sinusoidal disturbances in any direction is possible if and only if the transfer matrix $S(z) = I - G_0(z)K(z)$ has zero gain in all input directions at $z_i = e^{j\omega_i}$, for $i = 1, 2, \dots, n_f$. Then, the numerator of all elements of S must be zero at the frequencies of the disturbance. The condition $S(z_i) = I - G_0(z_i)K(z_i) = 0$ implies K is equal to the right inverse of G_0 at z_i 's which exists if and only if $G_0(z_i)$ is full row rank, i.e., $\text{rank}(G_0(z_i)) = n_y \leq u_u$. Moreover, from $S(z_i) = 0$, the numerator of the elements of $S(z)$ must contain the disturbance generating polynomial $\bar{D}_s(z) = \prod_{i=1}^{n_f} (z^2 - 2 \cos(\omega_i)z + 1)$ as a factor. So, we have $n_y n_u$ polynomial Diophantine equations which can be expressed in a matrix form. Let the monic polynomial $R_0(z)$ be the least common multiple of denominators of the elements of $G_0(z)$ and write

$$G_0(z) = \frac{1}{R_0(z)} [Z_{ij}(z)]_{n_y \times n_u}$$

$$K(z) = \frac{1}{z^N} [N_{ij}(z)]_{n_u \times n_y}$$

where the polynomials Z_{ij}, N_{ij} are the ij th elements of the above matrices, and $N_{ij}(z) = \theta_{ij0} + \theta_{ij1}z + \dots + \theta_{ij(N-1)}z^{N-1}$. The transfer matrix $S(z) = I - G_0(z)K(z)$ has zero gain in all directions if the numerator of each element of matrix $S(z)$ has $\bar{D}_s(z)$ as a factor, i.e.,

$$\bar{D}_s(z)L_{ii}(z) + \sum_{k=1}^{n_u} Z_{ik}N_{ki} = z^N R_0(z)$$

$$\bar{D}_s(z)L_{ij}(z) + \sum_{k=1}^{n_u} Z_{ik}N_{kj} = 0$$

for $i, j = 1, \dots, n_y, i \neq j$, and some polynomials L_{ij} . It is straightforward to show that (see [Kailath, 1980](#), Chapter 2) such equations can be written as an algebraic matrix equation $\mathbf{Ax} = \mathbf{b}$, where matrix $\mathbf{A}$ and vector $\mathbf{b}$ are known and depend on the coefficients of Z_{ij}^* 's, R_0 , and $\bar{D}_s$, $\mathbf{x} = [\theta^T, l^T]^T$, and θ, l are the concatenation of the unknown coefficients of N_{ij}^* 's, L_{ij} 's, respectively. The matrix $\mathbf{A}$ is full-row rank of dimension $n_u n_y (N + n_r + 1) \times n_u n_y (N + n_r + 1 + N - 2n_f)$, where $n_r = \deg(R_0)$. If $N \geq 2n_f$, the number of independent equations does not exceed the number of unknowns and this guarantees that the equation has at least one solution. If $N = 2n_f$, the matrix $\mathbf{A}$ is square and nonsingular and therefore the solution is unique. $\square$

For plants with fewer inputs than outputs, for degenerate plants $\text{rank}(G_0(z)) < \min\{n_y, n_u\}$, $\forall z$, and for plants with transmission zeros at the frequencies of the disturbance, complete rejection of disturbances in all directions is impossible. It should be noted that [Theorem 1](#) provides necessary and sufficient conditions for rejection of periodic disturbance signals in any direction; however, for a given disturbance vector, these conditions are sufficient as for a specific direction we just need $S = I - G_0K$ to have zero gain in the direction of the disturbance vector not necessarily in all directions. In a MIMO system, a signal vector u with frequency ω_0 can pass through $G_0(z)$ even if $G_0(z)$ has a transmission zero at this frequency; which is the case when the input vector u is not in the zero-gain directions of G_0 . The following two examples show that the conditions given in [Theorem 1](#) are not necessary for a specific direction of the disturbance vector.

Example 1 (A System with Zero Gain at a Single Frequency). Consider the following system (the sampling period is $T_s = 10^{-3}$ s):

$$G_0(z) = \frac{1}{z^2 - 0.5z + 0.5} \begin{bmatrix} z & 2 \cos(0.1)z - 1 \\ 1 & z \end{bmatrix},$$

which has a pair of zeros on the unit circle at $\omega_0 = 0.1$ rad/sample (100 rad/s) and the input direction corresponding to the zero is $\mathbf{v}_0 = [-1, e^{-0.1j}]^T$. From [Theorem 1](#), complete rejection of periodic disturbances with frequency ω_0 in all directions is impossible for this plant, because $\text{rank}(G_0(e^{j\omega_0})) = 1$ and the system has zero gain in some direction. However, for some disturbance vectors, say $\bar{d}_s(k) = [\sin(0.1k + 0.1), \sin(0.1k)]^T$, we can find a filter of the form (8) which completely reject $\bar{d}_s$, even though the conditions in [Theorem 1](#) are not satisfied. For example, for

$$K(z) = \frac{1}{z} \begin{bmatrix} 0.2469 & -0.4994 \\ 0.9906 & 0.2469 \end{bmatrix},$$

the transfer function matrix $S(z) = I - G_0(z)K(z)$ has zero gain at $\omega_0 = 0.1$ rad/sample in the direction of the disturbance vector $\bar{d}_s$, i.e., $\bar{d}(\omega_0) = [e^{0.1j}, 1]^T$, therefore the disturbance is rejected despite the fact that the conditions of [Theorem 1](#) are not satisfied ($\text{rank}(G_0(e^{j\omega_0})) = 1 < 2$ and $N = 1 < 2$).

Example 2 (A Degenerate System). Consider the following 2-input 2-output degenerate system (the sampling period $T_s = 10^{-3}$ s):

$$G_0(z) = \frac{1}{z} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

The largest possible rank of $G_0(z)$ is one, the minimum singular value of the system is zero for all frequencies, and the input direction corresponding to the zero of the system is $\mathbf{v}_0 = [1, -1]^T$. So, for this system we cannot reject periodic disturbances in every direction. However, for some disturbances, e.g., $\bar{d}_s(k) = [\sin(0.1k), \sin(0.1k)]^T$, there exists a filter of the form (8) for which $\bar{d}_s$ is completely rejected. For example, for

$$K(z) = \frac{0.74z - 0.4975}{z^2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix},$$

the transfer function matrix $S(z) = I - G_0(z)K(z)$ has zero gain at $\omega_0 = 0.1$ rad/sample (100 rad/s) in the direction of the disturbance vector $\bar{d}_s$, i.e., $\bar{d}(\omega_0) = [1, 1]^T$, and therefore the disturbance is rejected despite the fact that one of the conditions of [Theorem 1](#) is not satisfied ($\text{rank}(G_0(e^{j\omega_0})) = 1 < 2$).

A disturbance rejecting parameter vector θ^* in the ideal case (i.e., when $v = 0$ and $\Delta_m = 0$) leads to a perfect performance and guarantees that the plant output converges to zero exponentially fast. However, in the presence of noise and modeling error, a large gain of $G_0(z)K(z, \theta^*)$ may drastically amplify the noise part of

the disturbance. Moreover, it may significantly reduce the stability margin or lead to instability. The situation may get worse if the disturbance has some modes $z = e^{j\omega_i}$ near the transmission zeros of $G_0(z)$. In such case, $G_0(z)$ is close to becoming rank deficient at the frequencies of the disturbance ω_i 's. Indeed, in order to cancel the periodic disturbance terms the controller has to generate periodic terms of the same frequencies which after going through $G_0(z)$ result in periodic terms identical to that of the disturbance but of opposite sign. Clearly, when the plant has a very low gain at the frequencies of the disturbance, the filter gain must be large enough to make the gain of G_0K in the direction of $\bar{d}_s$ close to one. This however may increase $\|G_0K\|_\infty$ leading to noise amplification.

Lemma 1. If the conditions in [Theorem 1](#) are satisfied and $N > 2n_f$, there exists a θ^* with $\|\theta^*\| \leq r_0$, for some $r_0 > 0$, for which the effect of periodic components of the disturbance on the plant output is completely rejected and

$$J(\theta) = \|G_0(z)K(z, \theta)\|_\infty,$$

$$\text{s.t. } \theta \in S_\theta \triangleq \{\theta \mid \|\theta\| \leq r_0, \exists l: \mathbf{A}\mathbf{x} = \mathbf{b}\}$$

is minimized, where $\mathbf{x} = [\theta^T, l^T]^T$, and $\mathbf{A}, \mathbf{b}$ are defined in the proof of [Theorem 1](#).

Proof. If the conditions in [Theorem 1](#) are satisfied, the set S_θ is nonempty and affine, closed and convex. The function $J(\theta)$ is continuous and convex and attains its minimum on any compact set ([Bertsekas, Nedic, & Ozdaglar, 2003](#)). The above minimization problem can be solved by using linear matrix inequality (LMI) based solvers ([Boyd, Ghaoui, Feron, & Balakrishnan, 1994](#); [Nesterov & Nemirovski, 1994](#); [Varga & Parrilo, 2001](#)). $\square$

To design a filter with a satisfactory performance one may use either or both of the following two remedies:

- (1) Increasing the order of the filter: Increasing the value of N provides more flexibility in selecting the best parameter vector θ^* for which the periodic components of the disturbance are rejected and the $\mathcal{H}_\infty$ -norm of sensitivity function of the output with respect to the random noise in the disturbance is minimized.
- (2) Pre-filtering modification: If the plant has a small gain at the frequencies of the disturbance and comes close to lose rank at these frequencies, we may have a very poor performance because of possible large value of G_0K at other frequencies which may lead to noise amplification. Moreover, in the presence of high-frequency unmodeled dynamics if the plant has relatively large gain at high frequencies, with a periodic-disturbance-rejecting controller of the form (8), the closed-loop system may have a small stability margin or may become unstable. Since the plant model $G_0(z)$ is known, by a proper shaping of the singular value of $G_0(z)$ both performance and robust stability can be improved. That is, a pre-compensator $F_0(z)$ is designed such that $C'_0(z) = G_0(z)F_0(z)$ has a large enough gain (of the order of 1) over the frequency range of the disturbance in all directions, if possible, and has a sufficiently small gain at high frequencies. In the next section, we will discuss this modification and the trade offs between performance improvement and robustness with respect to plant modeling uncertainties.

The properties of the closed-loop system are summarized by the following theorem.

Theorem 2. Assume the conditions given in [Theorem 1](#) are satisfied. Let θ^* be the desired vector for rejecting the periodic disturbance terms while minimizing the sensitivity function of the output with

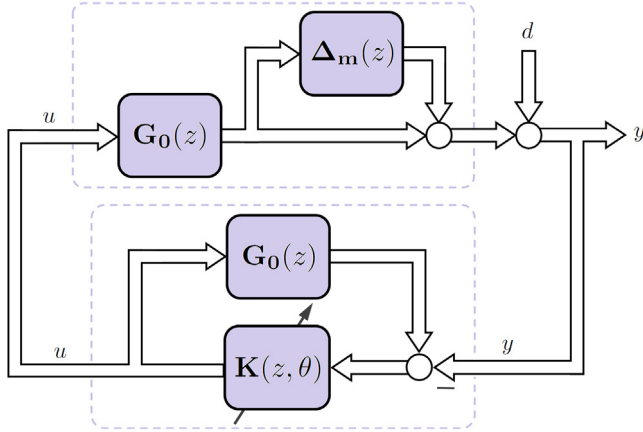


Fig. 3. The structure of the adaptive closed-loop system.

respect to noise given in Lemma 1. If $\|\Delta_m(z)G_0(z)K(z, \theta^*)\|_\infty < 1$, then all signals in the closed-loop system are uniformly bounded and

$$\lim_{T \rightarrow \infty} \sup_{k \geq T} \|y(k)\|_\infty \leq \|(I - G_0K)(I + \Delta_m G_0K)^{-1}\|_1 v_0 \\ \leq c(1 + \|G_0K\|_1)v_0,$$

where $v_0 = \sup_k |v(k)|$ and c is a finite positive constant independent of noise level. In addition, in the absence of the noise, the plant output goes to zero exponentially fast with time.

Proof. From (4)–(6), and the small-gain theorem, if $\|\Delta_m(z)G_0(z)K(z, \theta^*)\|_\infty < 1$, then $\bar{d}(k)$ and therefore all signals in the closed-loop system are uniformly bounded. From (4) and (7) we can write $y(k) = S(z, \theta^*)[\bar{d}_s(k)] + S(z, \theta^*)[\bar{v}(k)]$, where $S(z, \theta^*) = I - G_0(z)K(z, \theta^*)$. Since $S(z, \theta^*)$ has zero gain at the frequencies of $\bar{d}_s(k)$ in all directions, then $y(k) = \epsilon_{\bar{d}_s}(k) + S(z, \theta^*)[\bar{v}(k)]$, where $\epsilon_{\bar{d}_s}(k) = S(z, \theta^*)[\bar{d}_s(k)]$ is exponentially decaying to zero signal. Using well-known results for LTI systems (Desoer & Vidyasagar, 1975; Ioannou & Sun, 1996; Skogestad & Postlethwaite, 2005), the steady-state bound for the output $y(k)$ is given by

$$\|y\|_\infty \leq \|(I - G_0K)(I + \Delta_m G_0K)^{-1}\|_1 \|v\|_\infty \\ \leq c(1 + \|G_0K\|_1)v_0$$

where $c = \|(I + \Delta_m G_0K)^{-1}\|_1$ is a finite positive constant. $\square$

The above theorem gives the factors that affect the performance of the proposed disturbance rejection controller. In the next section we combine the controller presented above in the case of known disturbance frequencies with a robust adaptive law in order to deal with the practical case where the disturbance frequencies are unknown.

5. Adaptive suppression of unknown periodic components

In order to counteract the effect of unknown disturbances on the plant output, an adaptive filter is required to adjust its parameters in the direction of minimizing the norm of the plant output. The structure of the closed-loop system with an adaptive filter is shown in Fig. 3, where the filter parameter vector $\theta(k)$ is calculated at each sample time.

Let θ^* be the unknown desired parameter vector which suppresses the output vector norm. If the conditions given in Theorem 1 are satisfied, there exists a desired θ^* which attenuates the effect of d on y and gives the best performance. By ‘best performance’ we mean complete rejection of periodic terms without amplifying the effect of broadband random noise.

From (8) and Fig. 3 we have the following relations:

$$u(k) = -K(z, \theta)[\zeta(k)], \\ \zeta(k) = y(k) - G_0(z)[u(k)], \\ K(z, \theta) = [\theta_{ij}^T \alpha(z)]_{n_u \times n_y}, \\ \alpha(z) = [z^{-N}, z^{1-N}, \dots, z^{-1}]^T. \quad (9)$$

In order to generate an online parameter estimator or adaptive law that would generate the estimate $\theta(k)$ of θ^* at each time instant k we express θ^* in an appropriate parametric model. The following lemma presents a parametric model for the closed-loop plant to be used for parameter estimation (Ioannou & Fidan, 2006; Ioannou & Sun, 1996).

Lemma 2. The closed-loop system (1), (9) can be parameterized as

$$\zeta = \Phi^T \theta^* + \eta, \quad (10)$$

where $\theta^* = [\theta_{11}^{*T}, \theta_{12}^{*T}, \dots, \theta_{n_u n_y}^{*T}]^T \in \mathbb{R}^{N n_u n_y \times 1}$ is the unknown desired parameter vector, $\zeta(k) = [\zeta_1(k), \dots, \zeta_{n_y}(k)]^T = y(k) - G_0(z)[u(k)]$ is measurable, and the regressor matrix is given by

$$\Phi^T(k) = G_0(z)[\bar{w}^T(k)] \quad (11)$$

where

$$\bar{w}(k) = \begin{bmatrix} w(k) & & \\ & \ddots & \\ & & w(k) \end{bmatrix}_{N n_u n_y \times n_u},$$

where $w(k) = [\alpha(z)^T[\zeta_1(k)], \dots, \alpha(z)^T[\zeta_{n_y}(k)]]^T \in \mathbb{R}^{N n_y \times 1}$, and the unknown error term $\eta(k)$ depends on the noise level and the size of unmodeled dynamics, and also depends on exponentially decaying to zero terms.

Proof. The proof is given in the Appendix. $\square$

It follows from Lemma 2 that in the ideal case when there is no noise and modeling error, the error term $\eta(k)$ is zero or goes to zero exponentially fast.

In (11), $\bar{w}(k)$ contains the frequency components of the disturbance vector and the regressor $\Phi(k)$ is obtained by passing $\bar{w}(k)$ through $G_0(z)$. If $G_0(z)$ has very low gains at some frequencies of the disturbance, it may severely attenuate those frequency components, so the regressor will carry almost no information on those frequencies with the consequence of making their identification and therefore their rejection difficult if at all possible in some cases. Moreover, the corresponding disturbance rejecting controller $K(z, \theta^*)$ will have a very high gain at the frequencies of the disturbance which can make the gain of $G_0(z)K(z, \theta^*)$ very large at other frequencies. Therefore, we need to shape the frequency response of the plant model by designing an LTI filter $F_0(z) = [F_{0ij}(z)]_{n_u \times n_u}$ analytic in $|z| \geq 1$ placed right after the adaptive filter $K(z, \theta)$ as shown in Fig. 4. The modified closed-loop system is then equivalent to Fig. 3 where $G_0(z)$ is replaced by $G'_0(z) = G_0(z)F_0(z)$. If partial knowledge on the frequency range of the disturbance is available, the filter $F_0(z)$ must be such that $G'_0(z)$ has large enough gains (of the order of 1) in all directions (if possible) over the expected disturbance frequency range. This not only increases the partial excitation level of the regressor, but also reduces $\|G'_0(z)K(z, \theta^*)\|_\infty$, thereby reducing the broadband noise amplification. On the other hand, to improve robustness with respect to unmodeled dynamics, the $\mathcal{H}_\infty$ -norm of $\Delta_m(z)G'_0(z)$ must be as small as possible at all frequencies. Since $\Delta_m(z)$ is small at low frequencies and could be large at high frequencies the filter $F_0(z)$ is chosen to make $G'_0(z)$ small at high frequencies in order to guarantee the norm of $\Delta_m(z)G'_0(z)$ is small at all frequencies. It

is unknown and we do not have a priori knowledge on the its norm, $\theta_{\max}$ can be a conservative bound in order to ensure that $\theta^* \in \Omega$. To prevent parameter drift one may use other modifications such as fixed σ -modification that requires no bounds on the set where the unknown parameters belong to; however, it destroys the ideal convergence properties of the adaptive algorithm (Ioannou & Sun, 1996).

To implement the projection in (14), P^{-1} is decomposed into $(P^{-\frac{1}{2}})^T(P^{-\frac{1}{2}})$ at each instant k to guarantee all properties of the corresponding recursive least-squares algorithm without projection:

$$\begin{aligned}\bar{\theta}(k) &= \theta(k-1) + P(k)\Phi(k)\varepsilon(k) \\ \bar{\rho}(k) &= P^{-1/2}(k)\bar{\theta}(k) \\ \rho(k) &= \begin{cases} \bar{\rho}(k) & \text{if } \bar{\rho}(k) \in \bar{\Omega} \\ \perp \text{ proj of } \bar{\rho}(k) \text{ on } \bar{\Omega} & \text{if } \bar{\rho}(k) \notin \bar{\Omega} \end{cases} \quad (16) \\ \theta(k) &= P^{1/2}(k)\rho(k),\end{aligned}$$

where the set Ω is transformed to $\bar{\Omega}$ such that if $\bar{\theta}(k) \in \Omega$, then $\bar{\rho}(k) \in \bar{\Omega}$. Since Ω is chosen to be convex and $\bar{\rho}(k) = P^{-1/2}(k)\bar{\theta}(k)$ is a linear transformation, then $\bar{\Omega}$ is also a convex set (Goodwin & Sin, 1984).

The control signal at time k is produced based on the most recent estimate of θ^* at this time, i.e., $\theta(k-1)$, then from Fig. 4 and (9) we have

$$u(k) = -F_0(z)K(z, \theta(k-1))[\zeta(k)]. \quad (17)$$

The following theorem summarizes the properties of the closed-loop system (1), (9)–(15).

Theorem 3. Consider the closed-loop system (1), (9)–(15). Then, if the modeling error term is such that

$$c_0 \|\Delta_m(z)G_0(z)F_0(z)\|_1 < 1 \quad (18)$$

where c_0 is a positive constant which depends on $\theta_{\max}$ and is independent of the size of unmodeled dynamics, then all signals in the closed-loop system are uniformly bounded. Moreover, the plant output satisfies

$$\lim_{T \rightarrow \infty} \sup \frac{1}{T} \sum_{i=k}^{k+T-1} \|y(i)\|_2^2 \leq c(v_0^2 + \mu_{\Delta_m}^2), \quad (19)$$

for any $k \geq 0$, where μ_{Δ_m} is positive constant proportional to the size of unmodeled dynamics, $v_0 = \sup_k |v(k)|$, and $c > 0$ is a finite constant independent of k and the size of modeling error and noise. In addition, in the absence of noise and unmodeled dynamics, the adaptive law guarantees the convergence of $y(k)$ to zero.

Proof. The proof is given in the Appendix. $\square$

It should be noted that the stability condition (18) is a sufficient condition for BIBO stability of the closed-loop system. Such a norm-bound condition states that the closed-loop system can tolerate nonzero but small-size modeling errors. By proper shaping of the singular values of the plant model (i.e., by an appropriate choice of filter F_0), the robustness with respect to modeling uncertainties can be improved and the system may tolerate relatively large-size modeling errors at high frequencies.

6. Simulation results

In this section we demonstrate the performance of the adaptive controller applied to a 2-input 2-output plant. We assume that the sampling period is $T_s = 10^{-3}$ s.

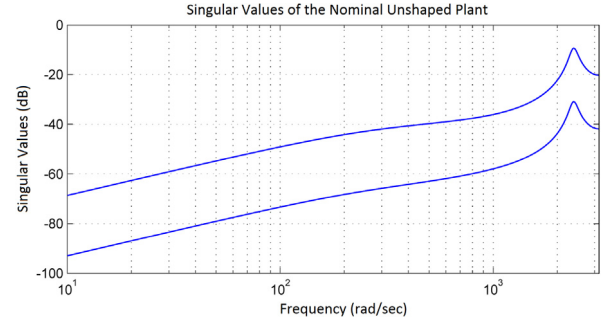


Fig. 5. The singular value plot of the unshaped nominal $G_0(z)$.

Consider the system

$$y = (I + \Delta_m(z))G_0(z)u + d$$

where $G_0(z)$ is the known modeled part given by

$$G_0(z) = \frac{0.01(z-1)}{R(z)} \begin{bmatrix} z-2 & z+0.5 \\ z-2 & z+1 \end{bmatrix}$$

where $R(z) = (z-0.75)(z^2+1.3z+0.8)$. For simulation purposes, the unknown multiplicative uncertainty term $\Delta_m(z)$ is assumed to be of the form

$$\Delta_m(z) = -\frac{10^{-5}}{(z+0.999)^2}I,$$

which has a negligible size at low frequencies and relatively large size near the Nyquist frequency. The unknown disturbance vector d at each sample time $t = kT_s$ is assumed to be

$$d(t) = \begin{bmatrix} d_{s1}(t) + v_1(t) \\ d_{s2}(t) + v_2(t) \end{bmatrix}$$

where

$$\begin{aligned}d_{s1}(t) &= 0.6 \sin(30t) + 0.6 \sin\left(95t + \frac{\pi}{8}\right) \\ &\quad + 0.6 \sin\left(180t + \frac{\pi}{6}\right), \\ d_{s2}(t) &= 0.5 \sin\left(20t + \frac{\pi}{4}\right) + 0.5 \sin\left(110t + \frac{\pi}{2}\right) \\ &\quad + 0.5 \sin\left(210t + \frac{\pi}{5}\right),\end{aligned}$$

and $v_1(t)$ and $v_2(t)$ are zero mean Gaussian with standard deviation 0.02 and $|v_i(t)| \leq 0.1$. We assume at $t = 25$ s new frequencies are added to the existing disturbance, i.e.,

$$\begin{aligned}d_{s1}(t) &= d_{s1}(t) + 0.9 \sin\left(103t + \frac{\pi}{7}\right) \\ d_{s2}(t) &= d_{s2}(t) + 0.9 \sin\left(128t + \frac{\pi}{3}\right),\end{aligned}$$

for $t \geq 25$ s. We also assume that at each time the number of distinct frequencies in the disturbance vector is at most 10 and we choose $N \geq 20$.

By using the adaptive control scheme proposed in Section 5, we generate a control signal u to counteract the effect of d on y . Fig. 5 shows the maximum and minimum singular values of the unshaped nominal $G_0(z)$. From the maximum singular value plot of $G_0(z)$ at high frequencies (around the Nyquist frequency π/T_s rad/s), we see that the closed-loop system seems to be vulnerable to modeling error in the presence of high-frequency unmodeled dynamics. From the minimum singular value plot, the plant has very small gains over the low frequency range which may adversely affect the performance.

In all simulations, we set $\gamma_0 = 1$ and $P^{-1}(0) = 0.01I$. The control signal $u(k)$ is applied at $t = 10$ s, and at $t = 25$ s new

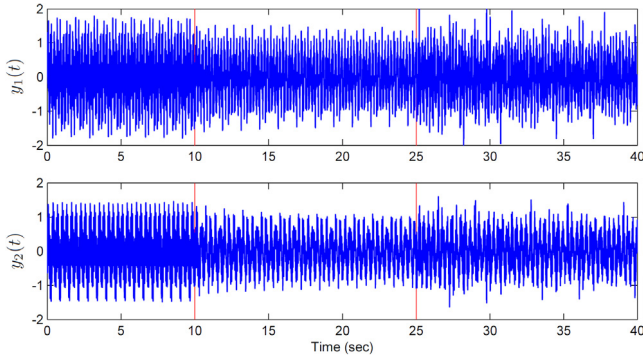


Fig. 6. The performance of the adaptive control system with $N = 20$ without pre-filtering modification. The control is applied at $t = 10$ s and at $t = 25$ s, new frequencies are added to the existing disturbance.

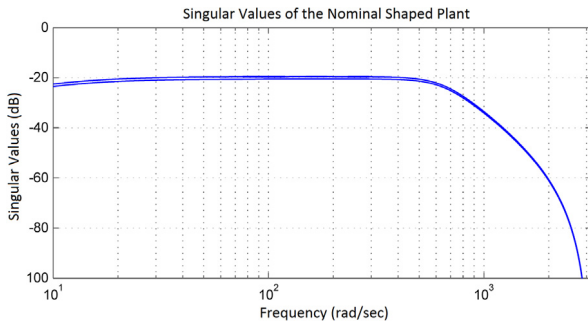


Fig. 7. The singular value plot of the shaped plant $G'_0(z) = G_0(z)F_0(z)$. The singular values are almost aligned.

frequencies are added abruptly to the disturbance. Fig. 6 shows the output of the adaptive closed-loop system with $N = 20$, without pre-filtering modification i.e., $F_0(z) = 1$. The poor performance is because of the low gain of the plant over the frequency range of the disturbance. We can increase the gain of the plant by multiplying it with a scalar gain and this will work fine in the absence of modeling errors. In the presence of modeling errors, which is the case in practice, increasing the plant gain using a constant scalar gain at all frequencies may lead to instability of the closed loop system.

It is clear from Fig. 6 that the lack of pre-filtering to shape the open-loop plant leads to a poor performance which is attributed to low plant gain at the frequencies of the disturbance. To improve the performance as well as robustness with respect to high-frequency unmodeled dynamics, we shape the singular values of $G_0(z)$ by an appropriate pre-compensation. To design a compensator $F_0(z)$ for this plant, we use the inner–outer factorization of plant nominal model. Since the plant has zeros on the unit circle, we apply the algorithm proposed in Chen et al. (2004) to the perturbed plant model $\tilde{G}_0(z)$ obtained by scaling the zeros by factor 0.99 to move them slightly away from the unit circle. Then $\tilde{G}_0(z) = \tilde{G}_{in}(z)\tilde{G}_{out}(z)$, where $\tilde{G}_{in}(z)$ is a stable proper all-pass filter and $\tilde{G}_{out}(z)$ is a stable proper with stable right inverse. Then, we choose $F_0(z) = \kappa_0 f_0(z)\tilde{G}_{out}^{-1}(z)$, where $\kappa_0 = 0.1$ and $f_0(z)$ is a scalar low-pass filter with DC gain of one designed to compensate the effect of high-frequency modeling uncertainties. We assume $f_0(z)$ is a third-order Butterworth low-pass filter with cutoff frequency of $0.2\pi/T_s = 630$ rad/s,

$$f_0(z) = \frac{0.018099(z+1)^3}{(z-0.5095)(z^2-1.251z+0.5457)}.$$

Then, $G'_0(z)$ is a low-pass filter with DC gain of $\kappa_0 = 0.1 = -20$ dB and cutoff frequency of 630 rad/s. The selection of the bandwidth of

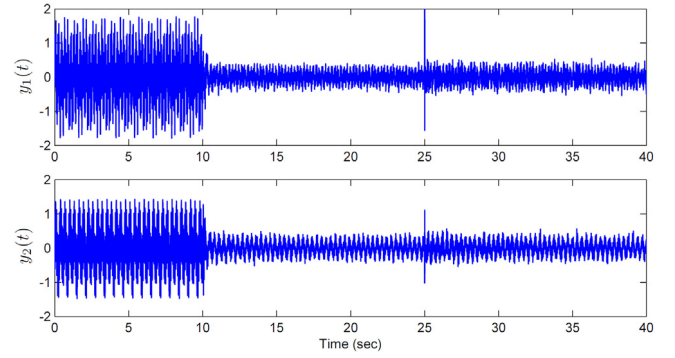


Fig. 8. The performance of the adaptive control system with $N = 20$ and pre-filtering modification.

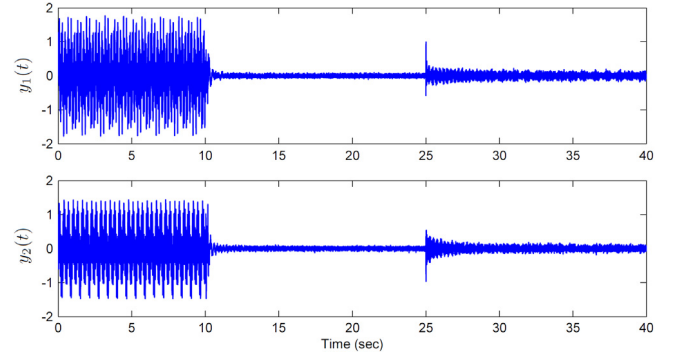


Fig. 9. The performance of the adaptive control system with larger adaptive filter order $N = 60$ together with pre-filtering modification.

$f_0(z)$ and the gain κ_0 is based on partial knowledge of the size and frequency range where the modeling error may be dominant. Large values of these two parameters can adversely affect the stability margin. The singular values of the shaped plant $G'_0(z) = G_0(z)F_0(z)$ are shown in Fig. 7.

Fig. 8 shows the performance with $N = 20$ and filtering modification. After closing the feedback loop at $t = 10$ s, the controller quickly adjusts its parameters to suppress the output norm by rejecting the sinusoidal part. At $t = 25$ s, the controller re-adjusts its parameters to counteract the effect of new periodic terms on the plant output. The performance of the proposed scheme with a higher order adaptive filter, $N = 60$, together with filtering modification is shown in Fig. 9. It is clear that both filtering and over-parametrization improve the performance of the closed-loop system considerably as it is predicted by analysis and demonstrated by Figs. 6, 8 and 9.

7. Conclusions

In this paper we address the problem of rejecting unknown periodic components of an output vector disturbance in the presence of output random noise and plant unmodeled dynamics for a MIMO plant that is stable and whose dominant part is known. We showed that by using proper plant pre-filtering, an over-parametrization for the number of unknown disturbance frequencies, and a robust adaptive law for parameter estimation we can achieve the following: (i) Guarantee stability provided the unmodeled dynamics are small in the low frequency range without amplifying the broadband output noise, (ii) Attenuation of unknown periodic components of the disturbance despite the presence of noise and unmodeled dynamics, and (iii) Improve performance especially in cases where the plant has very small gain at the frequencies of the disturbance.

Appendix A. Proof of Lemma 2

From (9) with $\theta = \theta^*$, the desired control input for which we can achieve the best performance can be written as

$$\begin{aligned} u^*(k) &= -K(z, \theta^*)[\zeta(k)] = -\Theta^* \Lambda(z)[\zeta(k)] \\ &= -\Theta^* w(k) \end{aligned}$$

where

$$\Theta^* = \begin{bmatrix} \theta_{11}^{*T} & \cdots & \theta_{1n_y}^{*T} \\ \vdots & \ddots & \vdots \\ \theta_{n_u 1}^{*T} & \cdots & \theta_{n_u n_y}^{*T} \end{bmatrix}_{n_u \times N n_y} \quad (\text{A.1})$$

$$\Lambda(z) = \begin{bmatrix} \alpha(z) & & \\ & \ddots & \\ & & \alpha(z) \end{bmatrix}_{N n_y \times n_y} \quad (\text{A.2})$$

and $w = \Lambda(z)[\zeta] = [w_1^T, \dots, w_{n_y}^T]^T$ and $w_i = \alpha(z)[\zeta_i]$. We can write the signal vector $\zeta(k)$ as

$$\begin{aligned} \zeta &= \zeta - G_0(z)[u^*] + G_0(z)[u^*] \\ &= -G_0(z)[u^*] + (\zeta + G_0(z)[u^*]) \\ &= -\xi^* + e^*, \end{aligned} \quad (\text{A.3})$$

where $e^* = \zeta + G_0(z)[u^*]$, and

$$\begin{aligned} \xi^* &= G_0(z)[u^*] = -G_0(z)[\Theta^* w] \\ &= -G_0(z) \begin{bmatrix} \sum_{j=1}^{n_y} \theta_{ij}^{*T} w_j \\ \vdots \\ \sum_{j=1}^{n_y} \theta_{n_u j}^{*T} w_j \end{bmatrix}. \end{aligned}$$

Then,

$$\zeta^T = \theta^{*T} \Phi + e^{*T} \quad (\text{A.4})$$

where $\theta^* = [\theta_{11}^{*T}, \theta_{12}^{*T}, \dots, \theta_{n_u n_y}^{*T}]^T$ is the unknown parameter vector and

$$\Phi = \begin{bmatrix} G_{011}(z)[w] & \cdots & G_{0n_y 1}(z)[w] \\ \vdots & \ddots & \vdots \\ G_{01n_u}(z)[w] & \cdots & G_{0n_y n_u}(z)[w] \end{bmatrix}$$

is the known regressor matrix. The transpose of Φ can be written as

$$\Phi^T = G_0(z) \begin{bmatrix} w & & \\ & \ddots & \\ & & w \end{bmatrix}^T.$$

The last term on the right hand side of (A.4) is

$$e^* = \zeta - G_0(z)[K(z, \theta^*)[\zeta]],$$

where θ^* is such that at each time $S(z, \theta^*) = I - G_0(z)K(z, \theta^*)$ has zero gain at the frequencies of the disturbance in all directions. The signal vector ζ can be written as $\zeta = d_s + v + \Delta_m G_0[u]$, then $|e^*|$ is proportional to the noise level and the size of unmodeled dynamics. Therefore, the parametric model of the system is given by

$$\zeta(k) = \Phi^T(k)\theta^* + \eta(k),$$

where the error term $\eta = e^*$, then we can write

$$\begin{aligned} \eta(k) &= S(z, \theta^*)[\zeta(k)] \\ &= S(z, \theta^*)[v(k) + \Delta_m(z)G_0(z)[u(k)]] + \epsilon_{d_s}(k) \end{aligned} \quad (\text{A.5})$$

where $\epsilon_{d_s}(k) = S(z, \theta^*)[d_s(k)]$ is exponentially decaying to zero term. $\square$

Appendix B. Proof of Theorem 3

From (8) and (17) the control input is given by

$$\begin{aligned} u(k) &= -F_0(z)K(z, \theta(k-1))[\zeta(k)] \\ &= -F_0(z)\Theta(k)\Lambda(z)[\zeta(k)], \end{aligned} \quad (\text{B.1})$$

where

$$\Theta(k) = \begin{bmatrix} \theta_{11}^T(k-1) & \cdots & \theta_{1n_y}^T(k-1) \\ \vdots & \ddots & \vdots \\ \theta_{n_u 1}^T(k-1) & \cdots & \theta_{n_u n_y}^T(k-1) \end{bmatrix}_{n_u \times N n_y}$$

and $\Lambda(z)$ is defined in (A.2). Then, from (B.1) and Fig. 3 we have

$$\zeta(k) = -\Delta_m(z)G'_0(z)[\Theta(k)\Lambda(z)[\zeta(k)]] + d(k),$$

then

$$\begin{aligned} \|\zeta_k\|_\infty &\leq \|\Delta_m(z)G'_0(z)\|_1 \|(\Theta\Lambda(z)[\zeta])_k\|_\infty + \|d_k\|_\infty \\ &\leq c_0 \|\Delta_m(z)G'_0(z)\|_1 \|\zeta_k\|_\infty + d_0, \end{aligned}$$

where $c_0 > 0$ is a finite constant whose boundedness is guaranteed due to projection, and d_0 is an upper bound for $\|d\|_\infty$. Then if $c_0 \|\Delta_m(z)G'_0(z)\|_1 < 1$, $\zeta(k)$ and then all signals in the closed-loop system are uniformly bounded.

With filter $F_0(z)$, the error term $\eta(k)$ in the parametric model can be expressed as

$$\eta(k) = S(z, \theta^*)[v(k)] + S(z, \theta^*)\Delta_m(z)G_0(z)[u(k)] \quad (\text{B.2})$$

where $S(z, \theta^*) = I - G'_0(z)K(z, \theta^*)$. In (B.2), the possible exponentially decaying terms are neglected as they do not affect the analysis.

Unless otherwise mentioned, for a signal vector $x(k)$, the notation $\|x(k)\|$ denotes the 2-norm of vector x at time k . For any vector $x(k) \in \mathbb{R}^n$, we have $\|x(k)\| \leq \sqrt{n}\|x(k)\|_\infty$. Then, using the boundedness of the signals in the closed-loop system we have

$$\begin{aligned} \|\eta(k)\|^2 &\leq 2(\|S[v(k)]\|^2 + \|S\Delta_m G_0[u(k)]\|^2) \\ &\leq 2n_y(\|S[v(k)]\|_\infty^2 + \|S\Delta_m G_0[u(k)]\|_\infty^2) \\ &\leq 2n_y(\|S[v]\|_k^2 + \|(S\Delta_m G_0[u])_k\|_\infty^2) \\ &\leq c(v_0^2 + \mu_{\Delta_m}^2) \end{aligned}$$

for some finite constant $c > 0$ independent of the size of v and $\Delta_m(z)$, where μ_{Δ_m} is positive constants proportional to the size of $\Delta_m(z)$, and $v_0 = \sup_k |v(k)|$. Since $m^2(k) \geq 1$, we can write

$$\left\| \frac{\eta(k)}{m(k)} \right\|^2 \leq c(v_0^2 + \mu_{\Delta_m}^2) \triangleq \bar{\mu}. \quad (\text{B.3})$$

Since $P^{-1}(0) = P^{-T}(0) > 0$, from (14) we have

$$P^{-1}(k) - P^{-1}(k-1) = \frac{\Phi(k)\Phi^T(k)}{m^2(k)} \geq 0 \quad (\text{B.4})$$

then $P^{-1}(k) = P^{-T}(k) > 0$ and with covariance resetting we have $P(k) = P^T(k) > 0$, for any k . Also, $P(k) - P(k-1) \leq 0$ implies $P(k) \leq P(k-1) \leq \dots \leq P(0)$, that is $P(k)$ is bounded. By pre-multiplying both sides of (B.4) by $P(k)$ and also post-multiplying it by $\tilde{\theta}(k-1) = \theta(k-1) - \theta^*$ we obtain

$$\begin{aligned} P(k)P^{-1}(k-1)\tilde{\theta}(k-1) &= \tilde{\theta}(k-1) \\ &\quad - P(k)\frac{\Phi(k)\Phi^T(k)}{m^2(k)}\tilde{\theta}(k-1). \end{aligned} \quad (\text{B.5})$$

From (10), (12), (13) we have

$$\Phi^T(k)\tilde{\theta}(k-1) = -m^2(k)\varepsilon(k) + \eta(k). \quad (\text{B.6})$$

Then, from (B.5), (B.6) we have

$$P(k)P^{-1}(k-1)\tilde{\theta}(k-1) = \tilde{\theta}(k-1) + P(k)\Phi(k)\varepsilon(k) - P(k)\frac{\Phi(k)\eta(k)}{m^2(k)}. \quad (\text{B.7})$$

Consider the following Lyapunov-like function whose boundedness is guaranteed due to projection and the boundedness of $P(k)$:

$$V(k) = \tilde{\theta}^T(k)P^{-1}(k)\tilde{\theta}(k),$$

where $\tilde{\theta}(k) = \bar{\theta}(k) - \theta^*$. If $\bar{\rho}(k) \in \bar{\Omega}$, $\tilde{\theta}(k) = \bar{\theta}(k)$, and from (16) $\bar{\theta}(k) = \bar{\theta}(k-1) + P(k)\Phi(k)\varepsilon(k)$, and from (B.7) we have

$$\tilde{\theta}(k) = P(k)P^{-1}(k-1)\tilde{\theta}(k-1) + P(k)\frac{\Phi(k)\eta(k)}{m^2(k)},$$

then

$$\begin{aligned} \Delta V(k) &\triangleq V(k) - V(k-1) \\ &= \tilde{\theta}^T(k)P^{-1}(k)\tilde{\theta}(k) - \tilde{\theta}^T(k-1)P^{-1}(k-1)\tilde{\theta}(k-1) \\ &= \tilde{\theta}^T(k-1)\left(P^{-1}(k-1)P(k)P^{-1}(k-1) - P^{-1}(k-1)\right)\tilde{\theta}(k-1) \\ &\quad + 2\tilde{\theta}^T(k-1)P^{-1}(k-1)P(k)\frac{\Phi(k)\eta(k)}{m^2(k)} \\ &\quad + \eta^T(k)\frac{\Phi^T(k)P(k)\Phi(k)}{m^4(k)}\eta(k). \end{aligned} \quad (\text{B.8})$$

From (B.4) and using the identity $(X + YZ)^{-1} = X^{-1} - X^{-1}Y(I + ZX^{-1}Y)^{-1}ZX^{-1}$ we can write

$$P(k) = P(k-1) - P(k-1)\frac{\Phi(k)Q\Phi^T(k)}{m^2(k)}P(k-1)$$

where $Q = Q^T = (I + \frac{\Phi^T(k)P(k-1)\Phi(k)}{m^2(k)})^{-1} > 0$. Then

$$P^{-1}(k-1)P(k)P^{-1}(k-1) - P^{-1}(k-1) = -\frac{\Phi(k)Q\Phi^T(k)}{m^2(k)}.$$

By simplifying (B.8) we obtain

$$\begin{aligned} \Delta V(k) &= -(m\varepsilon)^T Q (m\varepsilon) + (m\varepsilon)^T (2Q - 2I + 2R) \left(\frac{\eta}{m}\right) \\ &\quad + \left(\frac{\eta}{m}\right)^T \left(\frac{2I}{m^2} - \frac{Q}{m^2} - \frac{2R}{m^2} + R\right) \left(\frac{\eta}{m}\right), \end{aligned}$$

where $R = R^T = \frac{\Phi^T(k)P(k)\Phi(k)}{m^2(k)} \geq 0$. Then, from the boundedness of $\frac{\Phi(k)}{m(k)}$ and $P(k)$, it follows that

$$\begin{aligned} \Delta V(k) &\leq -c\|\varepsilon(k)m(k)\|^2 + c\|\varepsilon(k)m(k)\| \left\| \frac{\eta(k)}{m(k)} \right\| \\ &\quad + c \left\| \frac{\eta(k)}{m(k)} \right\|^2, \end{aligned} \quad (\text{B.9})$$

where c is a generic symbol for a positive constant. Using the inequality $-ax^2 + bx \leq -\frac{a}{2}x^2 + \frac{b^2}{2a}$ we have

$$\Delta V(k) \leq -c\|\varepsilon(k)m(k)\|^2 + c \left\| \frac{\eta(k)}{m(k)} \right\|^2. \quad (\text{B.10})$$

Therefore, from (B.3), (B.10)

$$\Delta V(k) \leq -c\|\varepsilon(k)m(k)\|^2 + c\bar{\mu}. \quad (\text{B.11})$$

Taking summation on both sides of (B.11) gives

$$\begin{aligned} \frac{1}{T} \sum_{k=1}^T \|\varepsilon(k)m(k)\|^2 &\leq c\bar{\mu} + \frac{c}{T}(V(0) - V(T)) \\ &\leq c\bar{\mu} + \frac{c}{T} \end{aligned}$$

that is $\varepsilon(k)m(k) \in \mathcal{S}(\bar{\mu})$ (i.e., it is $\bar{\mu}$ -small in the mean square sense). Since $m^2(k) \geq 1$, we have $\|\varepsilon(k)\|^2 \leq \|\varepsilon(k)m(k)\|^2$, therefore $\varepsilon(k) \in \mathcal{S}(\bar{\mu})$. Also,

$$\begin{aligned} \theta(k) - \theta(k-1) &= P(k)\Phi(k)\varepsilon(k) \\ &= P(k)\frac{\Phi(k)}{m(k)}\varepsilon(k)m(k) \end{aligned}$$

then

$$\begin{aligned} \|\theta(k) - \theta(k-1)\|^2 &\leq \lambda_{\max}^2(P(k)) \left\| \frac{\Phi(k)}{m(k)} \right\|^2 \|\varepsilon(k)m(k)\|^2 \\ &\leq c\|\varepsilon(k)m(k)\|^2 \end{aligned}$$

that is $\Delta\theta(k) \in \mathcal{S}(\bar{\mu})$.

To find the relationship between the plant output y and the estimation error ε , we write

$$\begin{aligned} y(k) &= G'_0(z)[u(k)] + \zeta(k) \\ &= -G'_0(z)[K(z, \theta(k-1))\zeta(k)] + \zeta(k). \end{aligned}$$

Let

$$\xi = G'_0(z)[u] = -G'_0(z)[\Theta w]$$

$$= -G'_0(z) \begin{bmatrix} \sum_{j=1}^{n_y} \theta_{1j}^T w_j \\ \vdots \\ \sum_{j=1}^{n_y} \theta_{n_{uj}}^T w_j \end{bmatrix}$$

where $w = \Lambda(z)[\zeta]$. Then, the i th element of ξ is given by

$$\xi_i = -G'_{0i1}(z) \left[\sum_{j=1}^{n_y} \theta_{1j}^T w_j \right] - \dots - G'_{0inu}(z) \left[\sum_{j=1}^{n_y} \theta_{n_{uj}}^T w_j \right].$$

From the Swapping Lemma, for a transfer function $G_{ij}(z)$ and vectors θ_{rs} and w_t we have

$$G_{ij}(z)[\theta_{rs}^T w_t] = \theta_{rs}^T G_{ij}(z)[w_t] + f_{ijt}(\Delta\theta_{rs}),$$

where $f_{ijt}(\Delta\theta_{rs}) = G_{ij}^c(z)[(G_{ij}^b(z)[w_t^T]\Delta\theta_{rs})]$, $\Delta\theta_{rs}(k) = \theta_{rs}(k+1) - \theta_{rs}(k)$, and $G_{ij}^b(z)$ and $G_{ij}^c(z)$ are defined in Section 2. Then,

$$\begin{aligned} \xi_i &= -\sum_{j=1}^{n_y} \theta_{1j}^T G'_{0i1}(z)[w_j] - \dots - \sum_{j=1}^{n_y} \theta_{n_{uj}}^T G'_{0inu}(z)[w_j] \\ &\quad - \sum_{j=1}^{n_y} f_{i1j}(\Delta\theta_{1j}) - \dots - \sum_{j=1}^{n_y} f_{inu_j}(\Delta\theta_{n_{uj}}). \end{aligned}$$

Then, by using (10) we can write

$$y(k) = -\Phi^T(k)\tilde{\theta}(k-1) + \eta(k) + f(\Delta\theta(k-1)),$$

where $f(\Delta\theta(k-1))$ is a function of $\Delta\theta(k-1) = \theta(k) - \theta(k-1)$ as defined above. Then from (B.6),

$$\varepsilon(k)m(k) = \frac{y(k) + f(\Delta\theta(k-1))}{m(k)}.$$

Since $m(k)$ is bounded and $\varepsilon(k)m(k)$, $\Delta\theta(k) \in \mathcal{S}(\bar{\mu})$, then $y(k) \in \mathcal{S}(\bar{\mu})$ which can be written as (19).

If $\bar{\rho}(k) \notin \bar{\Omega}$, then we have $V(k) = (\bar{\rho}' - \rho^*)^T (\bar{\rho}' - \rho^*)$, where $\bar{\rho}'(k)$ is an orthogonal projection of $\bar{\rho}(k)$ on $\bar{\Omega}$. Then, since $\rho^* \in \bar{\Omega}$, $(\bar{\rho}' - \rho^*)^T (\bar{\rho}' - \rho^*) \leq (\bar{\rho} - \rho^*)^T (\bar{\rho} - \rho^*) = \tilde{\theta}^T(k)P^{-1}(k)\tilde{\theta}(k)$. This enables us to establish all properties of the adaptive law without projection (Goodwin & Sin, 1984). $\square$

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